Cell-type-specific compositional and transcriptional changes in mouse and human aging

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Abstract

We tested whether aging imposes one uniform transcriptional program across cell types. Two hypotheses made the idea testable. H2: the share of each cell type within an organ shifts in a conserved direction with age. H3: cells lose identity with age and transcribe more genes that do not belong to their program. We measured cell-type fractions, gene expression within specific cell types, and whole-profile distance from young references in mouse and human single-cell data, with exploratory arms in fly, worm, zebrafish, and naked mole rat. Fractions shifted only in specific cell types in specific organs, and the clearest human lung examples did not survive study-aware reanalysis as within-study age effects. A mouse-derived gene signature did not show age associations beyond expression-matched random signatures in MSSM or PsychAD. Whole expression profiles drifted away from young references in mouse but showed only a small overall change in human. Human brain analyses identified replicated gene changes in three interneuron subtypes amid broader age-associated expression change, without general drift of their whole profiles. Gene-level age effects shared direction across cell types more than a donor-level null allows in mouse. No statistically significant increase in off-program expression was detected in mouse lung at the tested depths, but confidence intervals were wide and biologically relevant increases were not excluded. Together, no uniform transcriptomic program was detected in the surveyed datasets. Upstream causes are not settled, and shared mechanisms with cell-type-specific consequences are not excluded.

Introduction

Two views of aging predict different transcriptomes in old tissue. In the first view, one upstream failure drives the rest, so old cells of every type should show the same broad pattern: weakened identity, transcription of genes that do not belong to their program, and movement away from the young state in a shared direction. In the second view, aging comprises time-associated processes that co-occur and interact with different weights in different tissues and cell types. That view predicts heterogeneity: some cell types change and others do not, directions differ, and no single pattern appears everywhere. Recent work has claimed universal transcriptomic hallmarks of mammalian aging [11]. Single-cell studies have reported cell-type-specific aging in worm [5], increased transcriptional variability in immune cells [15], and human aging signatures that differ by cell type in pancreas [16].

There is no agreed operational definition of aging in general. Intuitively, aging is loss of function after passage of time: tissues work less well as time passes, repair weakens, and the risk of disease and death rises. That intuition is not yet an operational definition, because it does not say what to measure. This paper therefore states its working operational definition plainly. In this study, aging is measured as three things: (a) cell-type fractions, the share of each cell type within an organ; (b) gene expression within specific cell types; and (c) the distance of a whole expression profile from a young reference. That is the paper's working definition for the analyses reported here, not a final word on aging itself. Any claim in this paper about aging is a claim about those three measurements, and any theory the paper constrains is constrained through them.

The word universal also needs an operational definition, because the headline claim depends on it. A uniform transcriptional program, as tested here, would have to satisfy four criteria: the same direction of

change in every cell type tested, not only in a subset; overlapping gene sets driving the change, not only correlated directions; comparable magnitude after scaling by species lifespan; and presence in each species tested, not only in one. Failing any one criterion does not prove that no shared upstream process exists. It shows only that no uniform program was detected at the transcriptomic level under these criteria. A shared upstream process whose consequences differ by cell type would fail these criteria by design, and it is not excluded by this paper.

Bulk tissue measurements cannot separate these views, because a bulk change can reflect a shift in which cells are present rather than a change inside any cell. Single cell and single nucleus atlases allow composition and within cell expression to be measured separately, provided that sampling depth, detection differences between studies, reference choice, and the large number of cell type tests are handled as part of the inference rather than assumed away [1, 2].

This study, Agescape, was designed around those separations. It asks whether a mouse derived transcriptional signature transports to human cohorts; whether the share of each cell type within an organ changes with age; whether complete expression profiles move away from young references; whether cells express more genes characteristic of other cell types as they age; and whether the targeted gene changes seen in human brain are global or specific. A formal concordance test asks a narrower and answerable version of the universality question: do gene level age effects share direction across cell types within a tissue or cohort more than permuted age labels allow? Exploratory analyses in fruit fly, worm, zebrafish, and naked mole rat are reported as supplementary material. They do not drive the title or the main claims.

The scope is stated at the outset. Every measurement here is transcriptomic. Chromatin, DNA damage, transposon RNA, protein, cell function, and actual cell turnover were not measured. The results therefore test downstream transcriptomic predictions of mechanistic theories. They do not test the upstream mechanisms themselves, and no claim in this paper depends on pretending otherwise.

Materials and Methods

Datasets and inferential units

The mouse atlas was Tabula Muris Senis droplet data distributed through CZ CELLxGENE [1]. Public human brain, kidney, and lung data came from CZ CELLxGENE Census release 2025-11-08 [2]. The human lifespan cortex arm was the PsychAD dorsolateral prefrontal cortex atlas (Synapse syn52396927 and pseudobulk resource syn62064518) [9]. Exploratory arms were naked mole rat skin (GEO GSE152886) [7, 8], the Aging Fly Cell Atlas (GEO GSE218661) [3], the Gao worm atlas (GEO GSE229022) [4], and zebrafish testis (GEO GSE275361) [6]. Table 1 summarizes the analysis cohorts, and Table 2 gives the public human dataset identifiers. Only primary, normal-disease human cells were included. Public human vector and composition analyses used donors aged at least 20 years. Donors, animals, fish, or pools were the inferential units, not individual cells. Depth is the total number of RNA molecules captured per cell, summarized as mean log UMI.

Table 1: Analysis cohorts. Cell counts are specific to the listed analysis and are not total atlas sizes.

Cohort	Units	Ages	Sexes	Tissues	Platform	Cells in analysis	Exclusions and contributing studies
Mouse, Tabula Muris Senis	23 mouse identifiers	1, 3, 18, 21, 24, and 30 months	17 male, 6 female	16 tissues	10x droplet single-cell RNA-seq	238,186 cleaned	TMS; gene, UMI, and doublet QC; pooled identifiers evaluated in sensitivity analyses
Human public cohorts	621 signature-scoring donors; 479 vector donors	2 to 89 years; adult subsets used for vectors and composition	Male and female	Brain, kidney, lung	10x, Drop-seq, and Seq-Well	3,722,704 signature-scored	Eleven Census datasets; primary normal-disease cells; analysis-specific donor and cell floors

Cohort	Units	Ages	Sexes	Tissues	Platform	Cells in analysis	Exclusions and contributing studies
Human PsychAD	284 control donors	0.2 to 97 years	Male and female	Dorsolateral prefrontal cortex	Single-nucleus RNA-seq	1,307,674 nuclei	MSSM and HBCC banks; noncontrol donors excluded; 27 cell types
Naked mole rat	5 animals	1.1 and 11.3 years	Not verified	Skin	Single-cell RNA-seq	10,205 eligible	Savina and colleagues; sample 2V excluded; eligible basal and corneum cells
Fruit fly, AFCA	72 pools: 36 head and 36 body	30, 50, and 70 days	Male and female	Head and body	Single-nucleus RNA-seq	295,543 annotated	AFCA; day-5 cells and unannotated cells excluded
Worm, Gao atlas	12 pools	1, 6, 12, and 14 days	Not used as a covariate	Whole animal	Single-cell RNA-seq; NovaSeq 6000 and NextSeq 2000	81,196 annotated	Wild-type N2 fed OP50; singlets; singleton annotation excluded
Zebrafish	10 fish broad; 6 fish fine	5 to 27 months broad; 5 to 22 months fine	Male	Testis	Single-cell RNA-seq	69,576 broad; 36,523 fine	Sposato and colleagues; technical channels aggregated by fish

Table 2: Public human cohorts used for signature scoring. Donor, age, sex, and cell counts describe the scored control set. Later analyses used their prespecified eligible subsets.

Cohort	Census dataset identifier	Donors	Ages, years	Sex, M/F	Cells	Assays
MSSM DLPFC	`37a17b78-4864-4a42-b67b-31c00962795a`	213	19 to 89	131/82	842,777	10x 3' v3
HBCC DLPFC	`5c97eeeb-7e52-44b3-b010-b832b1f5424d`	134	2 to 85	85/49	688,828	10x 3' v3
RADC DLPFC	`5e57cd50-8e42-42d6-940d-5c1660d06864`	18	64 to 89	7/11	82,782	10x 3' v3
SEA-AD DLPFC	`6f7fd0f1-a2ed-4ff1-80d3-33dde731cbc3`	31	29 to 89	10/11	397,577	10x 3' v3; 10x multiome
Kidney Normal Adult	`0b4a15a7-4e9e-4555-9733-2423e5c66469`	18	43 to 74	13/5	48,783	10x 3' v3
Kidney Living Donor	`0bae7ebf-eb54-46a6-be9a-3461cecefa4d`	9	33 to 65	9/10	27,568	10x 3' v3
Kidney SN	`a12ccb9b-4fbe-457d-8590-ac78053259ef`	10	14 to 76	4/6	62,655	10x 3' v3
Kidney SCR	`dea717d4-7bc0-4e46-950f-fd7e1cc8df7d`	18	30 to 66	7/11	25,317	10x 3' v3
Lung CellRef	`eb499fd8-7000-419f-8854-926b9b61b11f`	21	12 to 76	10/11	122,445	10x 3' v2 and v3
Lung Integrated	`9f222629-9e39-47d0-b83f-e08d610c7479`	337	2 to 81	83/54	597,855	10x 3' and 5'; Drop-seq; Seq-Well
Immune Aging BAL	`1b350d0a-4535-4879-beb6-1142f3f94947`	12	20 to 75	6/6	826,117	10x 5' v2 and 3' v3

Mouse processing and differential expression

Mouse cells were filtered using the prespecified atlas quality control rules: 500 to 5,000 detected genes, at least 1,000 UMIs, and doublet removal. Mitochondrial filtering was not applicable because the released droplet object lacked mitochondrial transcripts. Counts were normalized to 10,000 per cell and transformed with log1p, using SCANPY [21]. For differential expression, donor by cell type pseudobulk counts were aggregated and log transformed after counts per million scaling. Age effects were estimated within each cell type with sex included when identifiable. P values used t based regression inference and Benjamini-Hochberg correction within the stated family [24]. Per-gene calls were checked for method sensitivity: the primary calls used DESeq2, and moderated weighted least squares was run as a sensitivity analysis. The two methods give very different call counts, as reported in the Results, so no per-gene conclusion in this paper rests on one method alone.

Signature derivation and transport

P1_Clean contained 139 mouse genes, 89 up and 50 down. P0_Raw257 contained 257 genes and was retained as a sensitivity score. P1 used uniform Benjamini-Hochberg differential expression candidates from adequately covered mouse units and removed predicted or pseudogene labels, non protein coding

biotypes, ribosomal and mitochondrial genes, and specified ambient RNA cases. Sex dimorphic and immediate early filters belonged to P2 and P3, not P1. One to one mouse to human orthologs came from the Ensembl derived mapping, giving 118 mapped P1 genes. The score was mean expression of up regulated genes minus mean expression of down regulated genes, with missing genes omitted rather than imputed. Public cohort regressions included age and sex when variable. Primary PsychAD models included age, sex, brain bank, postmortem interval, and three technical principal components. One Benjamini-Hochberg family covered 225 public cohort cell type tests and 27 primary PsychAD tests, for 252 tests. Matched null analyses in PsychAD and MSSM each generated 200 random signatures per cell type, matching the detected P1 up and down counts and expression-decile distribution, excluding P1 genes from the candidate pool, and using the same up-minus-down score. PsychAD regressions used the fully adjusted model; MSSM regressions used age and sex. Empirical p values were centered on the random-signature mean and corrected separately across 27 PsychAD and 21 MSSM cell types.

Cell type composition

A cell fraction is the share of one specific cell type among recovered cells in one organ, modeled as a centered log ratio slope. A fraction is never a tally across different denominators, never an absolute cell count, and a change in a fraction is not by itself evidence of fibrosis or of organ damage.

For donor i and cell type j, counts were transformed with pseudocount a and the centered log ratio: CLR_ij = log((count_ij + a) / sum_j(count_ij + a)) minus the mean over j of the same logged shares. The primary pseudocount was 0.5. Primary regressions included age and sex when identifiable. Sensitivity models added log total recovered cells and evaluated pseudocounts 0.1, 0.5, and 1.0, and a global Benjamini-Hochberg recheck applied one correction across the core mouse, human, and naked mole rat tests. Propeller provided a donor level count model check: proportions were transformed as logit((count + 0.5) / rowSum(count + 0.5)), followed by limma empirical Bayes regression with age, log total cells, sex when variable, and platform for Gao worm [19, 20]. Recovered composition was not interpreted as absolute tissue cell number. Study-aware models for human lung added study fixed effects, per-study slopes, and fixed and random effects meta-combination across studies, as reported in the Results.

Whole expression vector displacement

An expression vector is a complete within-cell expression proportion vector. No UMI downsampling was used for the direct vector analysis. Mitochondrial, ribosomal, and sex linked marker genes were excluded using species appropriate symbols. Each cell raw count vector was converted to within cell proportions p, and a young reference probability vector q was constructed from pooled counts of the same annotated cell type, with young donors or animals excluded from their own reference. Hellinger distance was H(p,q) = sqrt(1 minus sum over g of sqrt(p_g q_g)). Each evaluated donor by cell type unit required at least 20 cells and at least 50 remaining young reference cells. Mouse young references used animals aged 1 or 3 months, and human references used donors aged 20 to 40 years. Continuous age models included sex and, in specified arms, mean log UMI. Species summaries used unit fixed intercepts and donor clustered standard errors. Effects were scaled to 10 percent of species maximum lifespan using AnAge values of 4 years for mouse, 122.5 years for human, and 31 years for naked mole rat [23]. The mouse leave one cell type out analysis removed each cell type across all tissues and refit the global model. Depth sensitivity arms fixed sequencing depth at 500, 1,500, and 3,000 UMIs per cell and at full depth, and a pseudobulk Hellinger arm aggregated cells before measuring distance. Young versus young controls compared young animals against a split young reference to test for spurious displacement. An 18 month reference arm tested whether drift continues past maturity.

Gene effect concordance across cell types

Donor pseudobulk counts were aggregated per cell type, normalized to counts per 10,000, and log1p transformed. Each cell type yielded a vector of gene-level age effects from models including age, sex when variable, mean log UMI, and platform for Gao worm. Pairwise Pearson correlations were computed between cell-type effect vectors within a tissue or cohort. Genes entered the primary analysis when detected in at least half of donor pseudobulks in at least half of the stratum's units. Species summaries weighted strata by their number of cell-type pairs. The joint donor-level null assigned one permuted age to each donor and shared it across all of that donor's cell types, preserving cohort, sex, and platform strata. Mouse, fly, worm, and zebrafish used 1,000 permutations. Naked mole rat used all ten assignments of two aged animals among five. Human concordance was not recomputed under the joint null, because the underlying data were not available locally; its value is shown for context only, with its p-value treated as invalid. This null preserves donor sharing across cell types.

Off program expression, entropy, and depth sensitivity

Off program expression means expression of genes that are uncommon in young cells of the target type but characteristic of another cell type. That outcome is never called entropy in this paper. A separate Shannon entropy measure is reported under its own name where it was computed.

Within-cell Shannon entropy was minus the sum over g of p_g log(p_g), after the stated gene exclusions. Off-program genes were uncommon in young target cells but characteristic of another lung cell type: detection in at most 5 percent of young target cells and at least 20 percent of another cell type, expression of at least 20 counts per 10,000, and at least fivefold enrichment over the target. Young references excluded the evaluated young animal. Animal means were regressed on age and sex with Benjamini-Hochberg correction across cell types. Analyses evaluated 500, 1,500, and 3,000 UMIs and full depth. The full-depth arm used no downsampling. All arms retained the 1,500-UMI inclusion threshold, fixed young-reference marker definitions across depths, and computed off-program fractions against actual total UMIs. The library-size arm added each animal's mean log actual pre-downsampling depth. Global models used cell-type intercepts and animal-clustered t-based inference.

Human nonlinear age models and annotation stability

Nonlinear age effects used a cubic B spline with three degrees of freedom, fixed before examining results. F tests compared nonlinear and linear models on the same donor by cell type units, with Benjamini-Hochberg correction within cohort separately for linear and nonlinear comparisons. For annotation stability, excluded and unassigned cells were tabulated by age and tissue or cohort, and Spearman tests assessed age associations of excluded and unassigned fractions. A fixed canonical marker panel was evaluated by donor by cell type log counts per 10,000 expression with age, sex, and depth covariates, requiring at least six donors and 20 cells per donor by cell type unit, with Benjamini-Hochberg correction across targeted marker tests. Marker change was not equated with identity loss.

Genome-wide human interneuron replication

MSSM and HBCC donor pseudobulk counts were summed within each annotated interneuron subtype. Primary cells were normal-disease cells with at least 1,500 UMIs, donor ages 20 to 120 years, and at least 20 cells per donor and cell type. Genes were retained when they had at least 10 total pseudobulk counts, detection in at least 10 percent of donors, and mean expression of at least one count per 10,000; technical genes were excluded. Log1p counts per 10,000 were fit with age, sex, and mean log UMI using precision-weighted least squares. Gene-specific variance was smoothed against mean expression, and donor weights reflected cell numbers. Benjamini-Hochberg correction was applied within each cohort and cell type. A sensitivity analysis moderated residual variances by empirical-Bayes shrinkage. The

same models were applied across 39 cohort-by-cell-type combinations to assess specificity. Replication analyses used genes tested in both cohorts. Primary statistics were genome-wide effect correlations and same-direction overlaps passing FDR in both cohorts. One thousand permutations shuffled HBCC labels within 30 coexpression blocks defined by dominant principal-component loadings, preserving coarse gene covariance. A broad GABAergic neuron analysis served as a comparator.

Statistical families and multiplicity

Each analysis family was defined before correction, and Benjamini-Hochberg correction [24] was applied within the stated family. Mouse differential expression was corrected within each tissue by cell type unit. Signature transport used one family of 252 tests across the public cohorts and PsychAD. Random-signature empirical p values were corrected separately across the 21 MSSM cell types and the 27 PsychAD cell types. Composition was corrected first within each organ or cohort and then across one global family containing the core mouse, human, and naked mole rat tests. Whole-expression vectors were corrected within each species and cohort. Concordance used one joint permutation test per species and scope. Off-program expression and entropy were corrected within the mouse lung depth sweep. Interneuron differential expression was corrected within each subtype and cohort, and coexpression-block permutation supplied one test per subtype. Nonlinear age models were corrected within cohort separately for linear and nonlinear comparisons. The annotation-stability marker panel formed one family of 187 tests. Non-significance was not treated as equivalence. No equivalence claim is made in this paper.

Reproducibility and reporting

Aggregate result tables were retained for every reported number. Controlled PsychAD donor level values and identifiers were not exported. Exploratory, secondary, and primary families were labeled explicitly.

Results

Cell fractions change locally, in specific types and organs

Fraction changes, defined as age slopes in the share of one cell type among recovered cells in one organ, occurred only in specific cell types and datasets. Under the global correction, seven mouse organs and two human cohort datasets retained at least one fraction association. MSSM retained three changes and integrated lung retained two (Figure 1).

Sensitivity analyses support interpreting these as fraction changes among recovered cells, not absolute tissue cell numbers. Models adjusted for log total recovered cells gave 15 significant mouse tests across six organs and ten human tests across three cohort datasets. Effects at pseudocounts 0.1 and 1.0 correlated at 0.9569 in mouse and 0.9434 in human. A propeller logit plus limma check agreed with the centered log ratio models, with Spearman rho 0.9738 in mouse and 0.9313 in human. Individual significance calls changed across models. No fraction result demonstrates fibrosis or organ destruction.

In the pooled integrated lung cohort [10] of 119 donors and 44 cell types, club-cell fractions declined with age (slope minus 0.060227, q = 0.004423), as did multiciliated epithelial fractions (slope minus 0.034154, q = 0.022513). With study fixed effects, the club slope was minus 0.0154 (p = 0.103) and the multiciliated slope was minus 0.0056 (p = 0.467). No study-specific club slope had p below 0.05 among the 10 estimable studies. Meta-combination gave a club fixed effect of minus 0.0159 (95 percent CI minus 0.0296 to minus 0.0022, p = 0.023) and random-effects estimate of minus 0.0158 (p = 0.036). The multiciliated fixed-effect estimate was plus 0.0098 (p = 0.095), and the random-effects estimate was plus 0.0075 (p = 0.487). The multiciliated finding was not supported, and the club association was substantially attenuated by study.

Study and sampling differences further limit interpretation. Recovered club fractions ranged from 0.00009 to 0.3075 across studies, and young and old studies sampled different anatomical sites. The

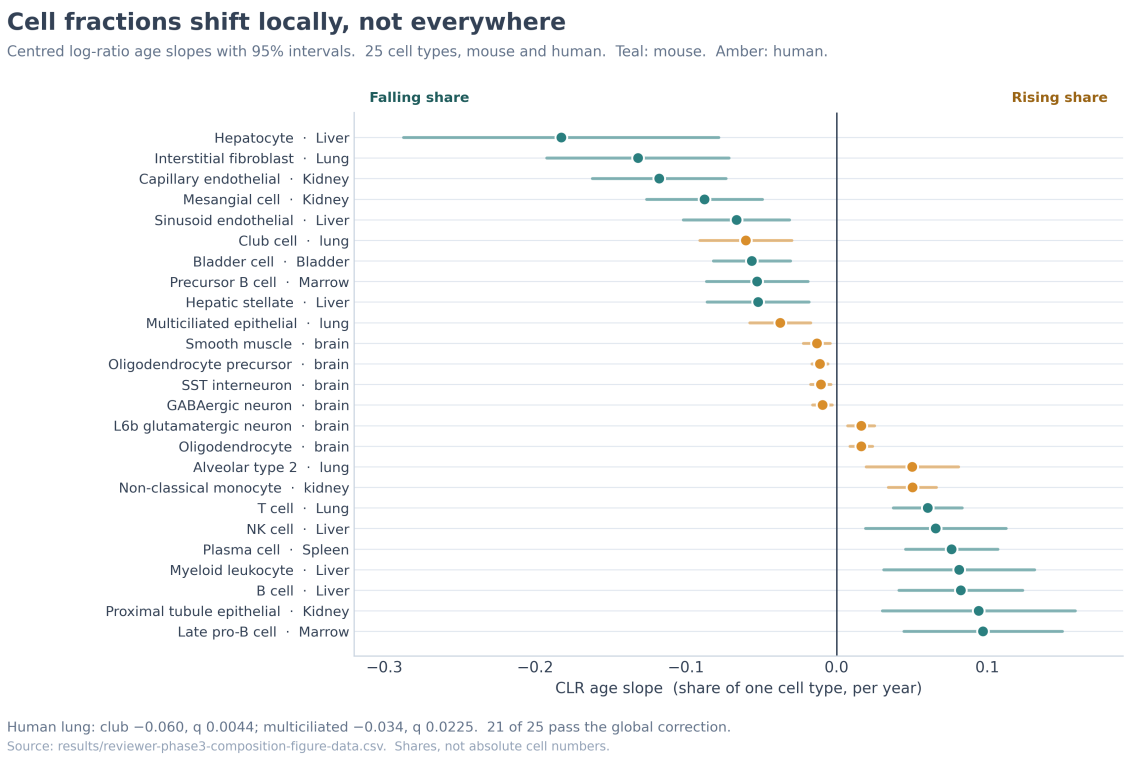


Figure 1: Figure 1. Cell-fraction age slopes with 95 percent confidence intervals. The 25 displayed results are the mouse and human associations passing FDR within their own strata at pseudocount 0.5, sorted by slope.

excluded-cell fraction increased from 16.1 percent at ages 20 to 39 to 34.6 percent at ages 60 to 81. Club-cell identity loss was not validly tested because a B-cell marker rule preceded the club-cell rule. A separate lung cohort of 17 donors had opposite-sign slopes in its nearest airway analogues. These pooled lung results are therefore between-study associations, not established within-study age effects.

A mouse-derived signature does not exceed matched random-signature nulls

The primary signature, P1, contains 139 mouse-derived genes, 89 up-regulated and 50 down-regulated with age, after technical and biotype filtering. The 257-gene P0 signature is a sensitivity analysis. P1 was derived from mouse differential expression before human transport evaluation, and no human result was used to choose its membership. One-to-one mouse-to-human mapping yields 118 genes, 76 up and 42 down, with missing genes omitted rather than imputed.

Signature derivation was method-sensitive. Moderated weighted least squares called 151 genes, DE-Seq2 called 312, and unmoderated weighted least squares called one across the same 17 mouse units. In leave-one-donor-out refits, 310 of 765 P1 gene-by-unit instances changed sign. P1 is therefore one method-dependent gene set, not a fixed biological program.

Three associations passed the 252-test global correction: MSSM layer 2/3 neurons ($q = 0.0208$), MSSM oligodendrocytes ($q = 0.0346$), and PsychAD EN_L3_5_IT_3 neurons ($q = 0.0208$). None survived the expression-matched random-signature analysis. In MSSM, the L2/3 neuron empirical p value was 0.0249 with null-family $q = 0.1045$, and the oligodendrocyte empirical p value was 0.004975 with $q = 0.0522$. No MSSM cell type had q below 0.05. No PsychAD cell type survived either; the smallest PsychAD q was 0.1007. Thus the signature did not transport beyond random signatures under this test, and the three global-screen associations are not presented as established transport.

PsychAD is not independent replication of the public cohorts. Exact identifier matching established that 103 of 214 public MSSM donors and 131 of 134 public HBCC donors also appear in the corresponding

PsychAD brain-bank arms, with concordant age and sex. The MSSM-to-HBCC interneuron comparison below uses different source banks.

Mouse whole profile displacement is distributed and survives depth adjustment, and the human overall displacement is small and imprecise

Mouse showed a robust global vector signal (an expression vector is a complete within-cell expression proportion vector, compared by Hellinger distance to a cross-fitted young reference from the same cell type, with no per-cell downsampling). The recomputed unadjusted effect is plus 0.1027 standard deviations per 10 percent of maximum lifespan, matching the published plus 0.1028 ($p = 0.00000398$). Centroid drift survives fixed sequencing depths: at 500 UMIs the linear depth-adjusted effect is plus 0.0953 ($p = 0.0139$), at 1,500 UMIs plus 0.1278 ($p = 0.0016$), at 3,000 UMIs plus 0.1490 ($p = 0.00039$), and at full depth plus 0.0700 ($p = 0.000015$). Dispersion, the spread of cells around their own type center, died at every fixed depth: minus 0.0017 at 500 UMIs ($p = 0.96$), plus 0.0145 at 1,500 UMIs ($p = 0.68$), and plus 0.0288 at 3,000 UMIs ($p = 0.41$). The mouse signal is therefore displacement of cell type centers away from young references, not increased scatter within cell types. Pseudobulk distance shows the same gradient: plus 0.1529 at full depth ($p = 0.0000024$), plus 0.0926 at 1,500 UMIs ($p = 0.00019$), and plus 0.0480 at 500 UMIs ($p = 0.073$, not significant). Young versus young controls pass, with permutation p values of 0.335 and 0.298, so the method does not manufacture displacement where no age difference exists.

Leave-one-cell-type-out global effects ranged from plus 0.0970 to plus 0.1119 around the full estimate of plus 0.1028, with a largest absolute change of 0.00912 standard deviations. The largest reductions followed removal of mesenchymal stem cells, podocytes, fenestrated cells, lung fibroblasts, or endothelial cells. No single cell type removed the global association (Figure 2). A donor-cluster bootstrap over the 23 mice gave a mean positive-unit fraction of 0.676 (95 percent prediction interval 0.561 to 0.792), with probability 0.0065 of a fraction at or below 0.5. Leave-one-donor-out global effects ranged from plus 0.0903 to plus 0.1088.

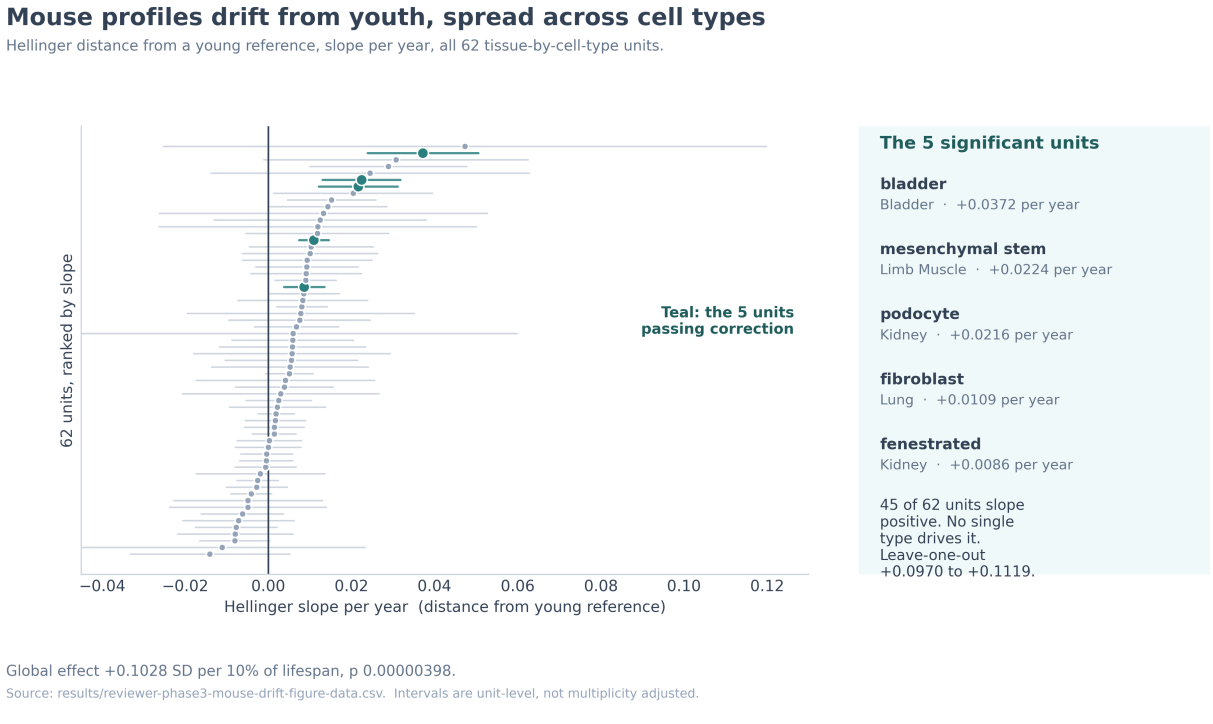


Figure 2: Figure 2. Age slopes in mean Hellinger distance from cross-fitted young references for all 62 mouse tissue-by-cell-type units. Bars are 95 percent confidence intervals.

With an 18-month reference, the depth-adjusted linear effect was plus 0.0072 ($p = 0.67$), and pseudobulk effects ranged from minus 0.010 to minus 0.018. The displacement was concentrated in the young-to-mature transition rather than continuing past 18 months.

Per-organ estimates were uneven. Kidney was robust across depths. Lung was significant at full depth (plus 0.0850, $p = 0.0065$) and at 3,000 UMIs (plus 0.155, $p = 0.030$). Tongue was plus 0.149 ($p = 0.0046$) and bladder plus 0.131 ($p = 0.048$) at full depth. Spleen was significant only at 1,500 and 3,000 UMIs. No association was detected in marrow at any depth, and heart and aorta estimates were negative but not significant.

The human overall vector effect was plus 0.0365 standard deviations ($p = 0.2218$), with a 95 percent confidence interval of minus 0.0221 to plus 0.0951. That interval cannot distinguish no effect from a mouse-sized effect: the mouse point estimate of plus 0.1028 sits just 0.0077 standard deviations above the human interval upper bound. No equivalence claim is made. The mouse minus human interaction is minus 0.0663 unadjusted (95 percent confidence interval minus 0.1330 to plus 0.0004, $p = 0.0515$, bootstrap $p = 0.072$) and minus 0.0447 after depth adjustment (95 percent confidence interval minus 0.0760 to minus 0.0135, $p = 0.0051$, bootstrap $p = 0.030$). On a range scale the interaction is not significant (unadjusted $p = 0.219$, adjusted $p = 0.057$). Human vector effects are never robust to the depth model: linear plus 0.0231 ($p = 0.050$), inverse square root plus 0.0362 ($p = 0.013$), spline plus 0.0138 ($p = 0.20$). Accordingly, the human estimate is reported with its interval and is not interpreted as evidence of absence.

Gene level age effects share direction across cell types, without establishing a universal program

The concordance test compares gene-level age-effect vectors between cell types within a tissue or cohort. The joint null assigns one permuted age per donor across all contributed cell types, preserving cohort, sex, and platform strata. Mouse within-tissue concordance with Scrublet doublet removal was plus 0.1855 against a null mean of plus 0.1378 (null 95 percent interval plus 0.0973 to plus 0.1947; $p = 0.0440$) across 78 units and 22 mice. Cross-tissue concordance was plus 0.0362 against plus 0.0231 ($p = 0.153$), so excess concordance was restricted to within-tissue comparisons.

Human concordance was plus 0.1891 against a null mean of plus 0.2076 (null 95 percent interval plus 0.1548 to plus 0.2702; $p = 0.7003$) across 131 units and 480 donors in seven cohorts. No excess human concordance was detected. In fruit fly, concordance was plus 0.2191 against plus 0.0384 ($p = 0.001$) across 99 units and 72 pools. In worm, it was plus 0.3718 against plus 0.1841 ($p = 0.003$) across 13 units and 12 pools. In zebrafish, it was plus 0.3415 against plus 0.3254 ($p = 0.361$) across 14 units and six fish. Naked mole rat concordance was plus 0.6146 against plus 0.6423 (exact $p = 0.70$) in five animals. Thus fly and worm showed excess concordance, whereas human, zebrafish, and naked mole rat did not (Figure 3).

Shared direction of gene-level effects is not evidence that every gene changes, that whole-vector distance increases, or that the same genes respond across species. A conserved AP-1 immune aging signature has been reported previously [12], but concordance here does not establish a universal program.

Human interneurons show replicated genome-wide gene effects

Genome-wide MSSM-to-HBCC age-effect correlations were strong for VIP (Pearson $r = 0.7217$ across 1,853 genes), parvalbumin ($r = 0.6839$ across 1,790 genes), and somatostatin interneurons ($r = 0.7614$ across 1,808 genes). Same-direction associations passing FDR in both cohorts numbered 76 VIP, 66 parvalbumin, and 249 somatostatin genes. Coexpression-block permutation gave $p = 0.001$ for each subtype, with null mean overlaps of 19.519, 17.382, and 80.442 genes, respectively. The broad GABAergic neuron comparator had a weaker genome-wide correlation ($r = 0.3315$), no same-direction both-FDR genes, and block-permutation $p = 1.0$ (Figure 4).

Empirical-Bayes moderation gave similar replicated overlaps: 77 VIP, 65 parvalbumin, and 239 somatostatin genes, again with block-permutation $p = 0.001$ for each subtype. Other cell types, including

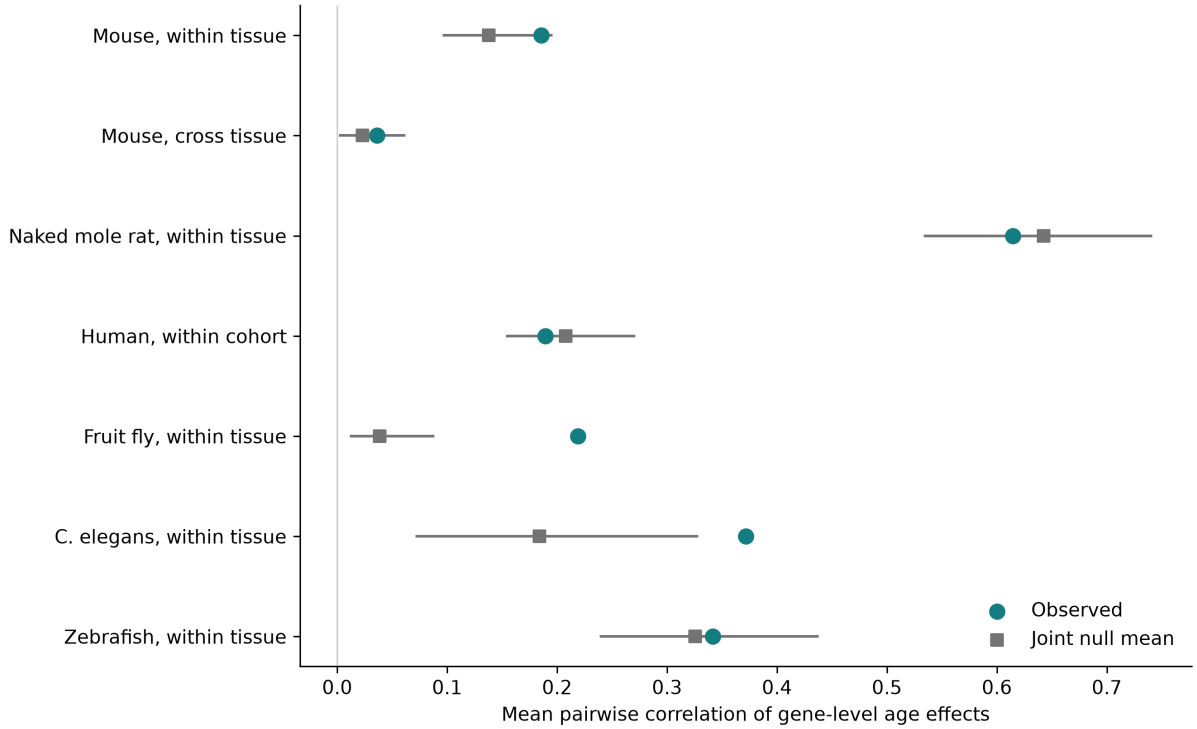


Figure 3: Figure 3. Observed mean pairwise Pearson correlations between cell-type gene-effect vectors (circles) and joint donor-permutation null means with 95 percent intervals (squares and bars).

microglia, oligodendrocytes, and excitatory neurons, also showed age-associated expression in the 39-combination analysis. Interneuron replication is therefore robust but not exclusive to interneurons.

Leave-one-donor-out refits of the 391 unmoderated replicated genes (76 VIP, 66 parvalbumin, and 249 somatostatin gene-by-cell-type entries; 782 gene-by-cohort refits) showed no slope changing sign, with a median largest single-donor shift of 0.48 standard errors; only three cases shifted by more than one standard error, and only four cases lost nominal significance in any refit.

Whole-vector drift was not detected after depth adjustment in these interneurons. The result is coordinated change in specific genes, not general displacement of their whole expression profiles. No protein, electrophysiological, or cognitive phenotype was measured.

Off-program expression in mouse lung: no significant increase detected, with wide intervals

The direct test of leaky, irrelevant gene transcription was run in mouse lung only. Off-program expression is the fraction of transcripts from genes that are uncommon in young cells of the target type but characteristic of another lung cell type. Analyses evaluated 500, 1,500, and 3,000 UMIs per cell and full depth, with young-reference marker definitions fixed across depths, plus a library-size covariate arm at full depth. Effects in the table are per 15 months.

Depth	Off-program effect	95 percent CI	p value	Entropy effect	95 percent CI	p value
500 UMIs	+0.2537	-0.2237 to +0.7311	0.2751	-0.0716	-0.4992 to +0.3559	0.7260
1,500 UMIs	+0.2438	-0.2349 to +0.7225	0.2948	-0.0789	-0.5135 to +0.3557	0.7041
3,000 UMIs	+0.1726	-0.3506 to +0.6958	0.4928	+0.0080	-0.4144 to +0.4304	0.9683
Full depth	+0.1743	-0.3557 to +0.7044	0.4940	-0.0906	-0.5517 to +0.3704	0.6812
Full depth, lib. size adj.	+0.1590	-0.3358 to +0.6539	0.5038	-0.0602	-0.3522 to +0.2318	0.6668

No statistically significant increase in off-program expression was detected in mouse lung at the tested depths, but confidence intervals were wide and biologically relevant increases were not excluded. No

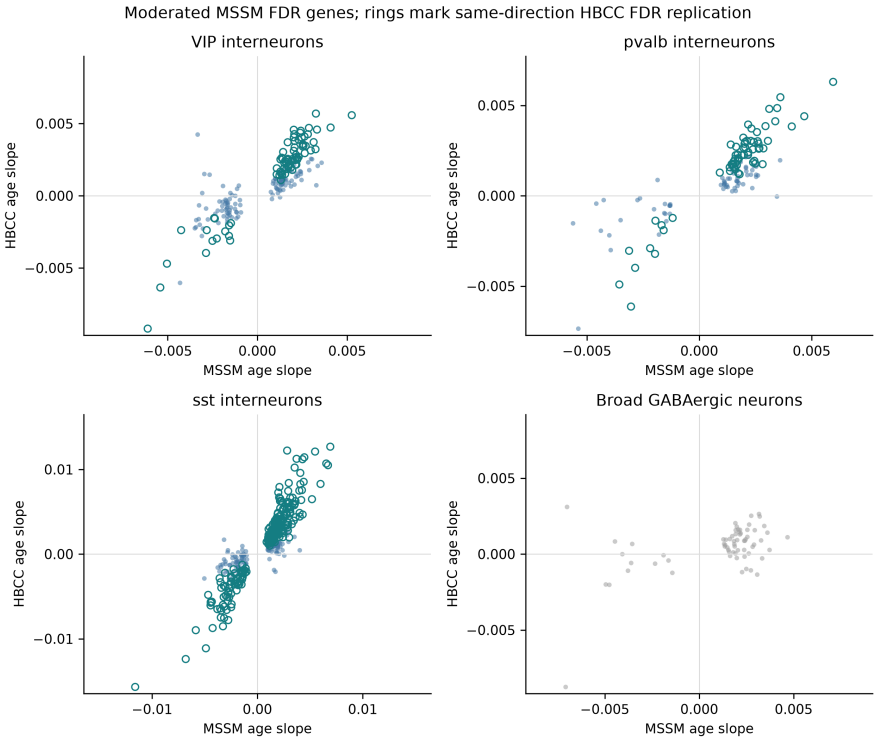


Figure 4: Figure 4. MSSM and HBCC age slopes for genes passing FDR in MSSM under empirical-Bayes moderation. Rings mark same-direction genes also passing FDR in HBCC. The broad GABAergic comparator is shown in grey.

depth produced a significant positive global off program association. The full depth null does not prove biological stability: the off program minimum detectable effect at 80 percent power was 0.6191 standard deviations, and equivalence was established at neither the 0.1 nor the 0.25 standard deviation margin. The entropy null is in the same position, with a minimum detectable effect of 0.5621 standard deviations and no equivalence at either margin.

The distinct cell-type-average displacement endpoint, measured without downsampling, gave an unadjusted Hellinger effect of plus 0.2914 standard deviations per 15 months (95 percent CI minus 0.1203 to plus 0.7030, p = 0.1653). After mean log UMI adjustment it was plus 0.2626 (95 percent CI plus 0.0806 to plus 0.4445, p = 0.00468). This endpoint is neither off-program expression nor entropy.

Adult only models, nonlinear age, and annotation stability

Every public human vector and composition donor was already at least 20 years old. The minimum age across both analyses was 20, so adult only refitting uses the same donor sets and changes nothing for these public analyses. Developmental donors in PsychAD belong to the separate signature transport analysis. Nonlinear models used a fixed three degree of freedom cubic B spline. MSSM had two linear false discovery rate hits and no nonlinear hits in the vector analysis, and six linear hits and one nonlinear hit in the composition analysis. Integrated lung had two linear composition hits and no nonlinear hits. These results should not be summarized as confirmation of linear age effects, because nonlinearity and cohort specific age coverage can change individual conclusions.

Annotation stability limits how far the composition results can be pushed. No per cell annotation confidence probability was available in the mouse or public human inputs, and doublet scores were not relabeled as annotation confidence. Age related differences in excluded or unassigned cells were detected in integrated human lung (Spearman rho = +0.4334, q = 0.0251 across 48 ages) and in mouse pancreas

(rho = +1.0 across four age groups, a result with only four groups that should not be generalized). A fixed canonical marker analysis found eight false discovery rate significant changes across 187 tests, seven in human and one in mouse, including increased microglial C1QA in MSSM (slope +0.00522 log counts per 10,000 per year, q = 0.000280) and decreased mouse cardiac fibroblast COL1A1 (slope -0.0397, q = 0.00617). These findings do not demonstrate identity loss, because marker expression can change without loss of cell identity. They do mean that identity loss cannot be excluded as an explanation for any affected composition result.

Exploratory and supplementary arms

These analyses were exploratory and did not drive the main claims. In fruit fly, the 38 depth-adjusted vector-hit cell types shared gene-level changes: 537 of 703 pairs shared at least one FDR-significant gene, with a median Jaccard index of 0.0130. Gene Ontology enrichment [22] was concentrated in specific units rather than one common pathway. Fly whole-vector effects were not detected in head (minus 0.142, p = 0.60) or body (plus 0.047, p = 0.86) at full depth. Gene-effect concordance under the joint null was plus 0.2191 against a null of plus 0.0384 (p = 0.001).

The worm vector result was reference-sensitive. A day-1 reference gave depth-adjusted effects of plus 1.59 to plus 1.68 standard deviations, whereas the balanced day-6 reference gave plus 0.5319 (parametric p = 0.0189; permutation p = 0.0535). Day 6 included both sequencing platforms. In zebrafish, the broad-compartment vector effect was plus 0.71 to plus 0.75, while the fine-grained effect was plus 0.2211 with a wide interval (95 percent CI minus 0.3588 to plus 0.8009) in six fish. Worm concordance also survives the joint null (plus 0.3718 against plus 0.1841, p = 0.003), while zebrafish concordance does not (plus 0.3415 against plus 0.3254, p = 0.361). In naked mole rat skin, the vector estimate was minus 0.4744 (exact p = 0.2) in five animals. These small or reference-sensitive arms do not support claims of stability, protection, or species differences. Naked mole rat mortality context [18] does not provide evidence for a transcriptomic effect.

Discussion

The measured pattern was cell-type-specific. Fraction associations were local; mouse whole-profile displacement was distributed but concentrated in the young-to-mature transition; the overall human displacement estimate was imprecise; and human brain analyses showed replicated gene-level change in three interneuron subtypes amid broader expression change. The mouse-derived P1 signature did not exceed expression-matched random-signature nulls. No uniform transcriptomic program was detected in the surveyed datasets. Shared upstream mechanisms with cell-type-specific consequences remain possible. Within-tissue concordance in mouse, fly, and worm adds a narrower coordination result: some gene-level age effects share direction across cell types within each species. Zebrafish does not show this. This is not equivalent to universal whole-profile deterioration, cross-tissue coordination, or shared genes across species.

Predictions

First, cell types with replicated RNA changes should show coupled regulatory changes in same-cell multiome measurements. Interneurons are a direct human test case, and mouse cell types with large young-to-mature displacement are a complementary test case. This prediction is motivated by epigenetic-information and reprogramming studies [13, 14], but the present data do not measure chromatin or assess upstream causes.

Second, genuine age-associated fraction changes should replicate across studies with the same anatomical sampling and donor-level inference. Study-matched lung sampling should clarify whether the club-cell association persists independently of study composition.

Third, larger depth-controlled cohorts should distinguish whether the wide off-program intervals conceal a meaningful increase. Such studies should report off-program expression, entropy, and whole-profile displacement as separate endpoints.

Limitations

These limitations are part of the results, not qualifications added after them.

The study is cross sectional and observational throughout. There is no perturbation, no longitudinal follow up, and no causal identification. Age associations can reflect cohort composition, technical variation, development, disease, or postmortem effects.

No equivalence claim is made in this paper. The human overall vector interval cannot distinguish no effect from a mouse-sized effect, and it is reported as imprecise rather than as evidence of absence. The off program, entropy, naked mole rat, and zebrafish nulls are likewise imprecise rather than evidence of stability.

Fly and worm samples are pools, not individually tracked animals, so within pool variation is not available to the inference. The naked mole rat arm has five animals, with skin only and group level ages, and the zebrafish fine grained arm is small. Neither can support claims of stability, protection, or species difference.

PsychAD and the public MSSM and HBCC cohorts contain exact identifier confirmed overlapping donors, so PsychAD analyses are not independent replication of the public cohorts. The MSSM to HBCC interneuron comparison uses different source banks.

Annotation is fixed while marker expression changes with age, and excluded cell fractions differ by age in at least two datasets. Identity loss therefore cannot be excluded as an explanation for any affected composition result.

Cell turnover was not measured. Published turnover estimates vary widely by cell type [17, 25], so no result is attributed to division history. No direct senescent-cell identification was performed; senescence gene sets [26] are not interchangeable with the off-program endpoint.

Depth, sequencing platform, and reference choice affect the measurements throughout. Platform and age are partly entangled in worm, and the worm vector effect changes size and security with the reference age.

Cross species effect sizes are standardized within each dataset. They are not directly comparable as measures of biological importance, because tissues, age windows, assays, annotation resolution, and lifespan coverage differ between species.

Conclusions

No uniform transcriptomic program was detected in the surveyed datasets. Fractions changed only in specific cell types in specific organs, whole-expression profiles drifted in mouse but showed a small and imprecise overall change in human, and the durable human findings included replicated gene changes in three interneuron subtypes amid broader expression change. The P1 signature did not exceed matched random-signature nulls. No statistically significant increase in off-program expression was detected in mouse lung at the tested depths, but confidence intervals were wide and biologically relevant increases were not excluded. Gene-level effects shared direction across cell types more than a donor-level null allows in mouse, fly, and worm, but not in zebrafish. Upstream causes are not settled, and shared mechanisms with cell-type-specific consequences are not excluded.

Data Availability

The analysis code and aggregate result tables are in the Agescape repository at https://github.com/Abolfazl-Alipour/agescape. Dataset accessions are given in the Materials and Methods. The results published here

are in whole or in part based on data obtained from the AD Knowledge Portal (NIH-NIA, U24AG061340). Controlled-access PsychAD donor-level values (Synapse syn52396927, pseudobulk resource syn62064518) are available from the source resource under its own terms and were not redistributed; only aggregate results are shared here.

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Competing Interests

The author declares no conflict of interest. [Pending author confirmation.]

Ethics Statement

This study performed no new human or animal work. All data analyzed are from published public cohorts and controlled access resources used under their own terms, as described in the Materials and Methods. No new human participants or animals were recruited, no interventions were performed, and no identifiable individual level data are reported. [Draft statement, pending author confirmation. No ethics approval number is claimed.]

Supporting Information

No supporting figures are included.

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